

Deep Neural Network for Lumbar Spondylolisthesis Diagnosis from X-ray Images: A Comparative Analysis of ResNet50_V2 and EfficientNetV2 Models

Md. Redwan Hossain

A Thesis in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT – International University of Business Agriculture and Technology

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Letter of Transmittal

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Subject: Letter of Transmittal.

Dear Sir,

As per the requirements of the thesis course, the advocates would like to present the proposal as per your instructions titled “Deep Neural Network for Lumbar Spondylolisthesis Diagnosis from X-ray Images: A Comparative Analysis of ResNet 50 V2 and EfficientNetV2 Models”.

The purpose of the paper is to detect ‘lumbar spondylolisthesis’ a lumbar spinal displacement disorder by using ‘EfficientNetV2’ and ‘ResNet 50 V2’ deep convolutional neural network and make a comparative study between these two model.

We hope that this request will merit your approval.

Yours sincerely,

Md.Redwan Hossain
20203005

Student's Declaration

This is to certify that the work presented in this thesis paper, titled, “Deep Neural Network for Lumbar Spondylolisthesis Diagnosis from X-ray Images: A Comparative Analysis of ResNet50_V2 and EfficientNetV2 Models”, is the outcome of the investigation and research carried out by the following student under the supervision of Suzad Mohammad, Lecturer, Department of Computer Science and Engineering, International University of Business Agriculture and Technology.

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20203005

Supervisor's Certification

This thesis paper titled “Deep Neural Network for Lumbar Spondylolisthesis Diagnosis from X-ray Images: A Comparative Analysis of ResNet50_V2 and EfficientNetV2 Models” submitted by the student as mentioned below has been accepted as satisfactory in partial fulfillment of the requirement for the degree of Bachelor of Computer Science and Engineering.

Suzad Mohammad

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Abstract

This thesis pioneers an advanced approach to lumbar spondylolisthesis diagnosis utilizing deep neural networks, specifically the ResNet50_V2 and EfficientNetV2 models. The study is conducted on a curated dataset consisting of 71 images representing normal lumbar spines and 79 images depicting spondylolisthesis conditions. Employing state-of-the-art deep learning architectures, our research explores the diagnostic potential of ResNet50_V2 and EfficientNetV2. The ResNet50_V2 model achieves an exceptional accuracy of 98.48%, surpassing the EfficientNetV2 model, which attains a commendable accuracy of 96.64%. This comparative analysis offers insights into the respective strengths and performance characteristics of these models in the context of lumbar spondylolisthesis detection. Furthermore, the investigation illuminates the nuances of diagnostic accuracy, precision, and recall metrics specific to each model. These discoveries not only add to the expanding realm of medical image analysis but also carry noteworthy implications for advancing the creation of precise and effective diagnostic instruments for lumbar spondylolisthesis.

Acknowledgments

I extend my heartfelt gratitude to all those who have been instrumental in the successful completion of this thesis on "Deep Neural Network for Lumbar Spondylolisthesis Diagnosis from X-ray Images: A Comparative Analysis of ResNet 50 V2 and EfficientNetV2 Models."

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I would like to acknowledge the participants who contributed their time and consent to the inclusion of X-ray images in the dataset. Their involvement has been pivotal in enhancing the relevance and applicability of this study.

This collaborative effort has been an enriching experience, and I am profoundly grateful to each individual who has played a role, no matter how small, in the realization of this thesis.

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Chapter 1. Introduction

1.1 Background and Context

Lumbar spondylolisthesis (LS) is the anterior shift of one of the lower vertebrae about the subjacent vertebrae. There are several symptoms to define LS, and these symptoms are not detected in the early stages of LS. This leads to disease progress further without being identified [1]. This condition can have symptoms of weakness that can be posterior pain, muscle pain, muscle pain, arthritis and precise timing to improve outcome and diagnosis is important.

Methods of categorizing and identifying diseases related to the vertebral column often involve the application of image-processing techniques. Image classification has remained a focal point of research, with Deep Learning (DL) methodologies offering extensive capabilities and adaptability within this domain [2]. Among the various types of Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs) have gained significant popularity. CNNs utilize multi-layered pixel-oriented Artificial Neural Network (ANN) methodologies [3]. Comprising an input layer, several hidden layers, and a solitary output layer, CNNs are extensively utilized for image classification. They have surpassed traditional feature-based methods in image classification tasks and have exhibited promising outcomes, particularly in the realm of medical imaging [4, 5].

Traditionally, the diagnosis of LS is mostly based on radiographic examination through X-ray imaging, where doctors look for spinal holes and other structural abnormalities but this procedure requires radiologist interpretive imaging. Besides MRI utilization is limited by its low accessibility. Spinal X-rays can rapidly provide information on the bony structure of the patient. However, this process involves subjective interpretation by radiologists, which can lead to variability in diagnostic accuracy and is susceptible to human error.

The emergence of deep learning techniques has revolutionized medical image analysis by providing automated and highly subjective diagnostic methods. In particular, Convolutional Neural Networks (CNNs) have shown extraordinary promise in various medical imaging tasks, including the detection and classification of pathological conditions from radiographs.

1.2 Problem Statement

Despite the continuous advancements in medical imaging technology, the accurate and timely diagnosis of lumbar spondylolisthesis remains a challenging task. Conventional diagnostic methods, often reliant on manual interpretation of X-ray images, may introduce subjectivity and variability in the diagnostic process. Moreover, the current literature lacks a comprehensive exploration of the comparative effectiveness of state-of-the-art deep learning models, such as ResNet 50 V2 and EfficientNetV2, specifically tailored for lumbar spondylolisthesis detection from X-ray images. This research seeks to address these gaps by developing, optimizing, and rigorously evaluating deep neural networks, ultimately contributing to the refinement of diagnostic tools for lumbar spondylolisthesis and advancing the integration of artificial intelligence in orthopedic diagnostics.

1.3 Objectives

- 1.3.1 **To Investigate the Efficacy of Deep Learning Models:** Evaluate the performance of ResNet50 V2 and EfficientNetV2 in diagnosing lumbar spondylolisthesis from X-ray images.
- 1.3.2 **To Compare the Accuracy of ResNet50 V2 and EfficientNetV2:** Conduct a comparative analysis of the diagnostic accuracy achieved by ResNet50 V2 and EfficientNetV2 models.
- 1.3.3 **To Explore the Interpretability of Model Predictions:** Investigate the interpretability of model predictions, seeking insights into the features or patterns driving the diagnosis of lumbar spondylolisthesis.
- 1.3.4 **To Examine the Impact of Dataset Characteristics:** Investigate how dataset characteristics, such as size, diversity, and annotation quality, influence the performance of ResNet50 V2 and EfficientNetV2 models.
- 1.3.5 **To Offer Recommendations for Clinical Implementation:** Provide

recommendations for the potential clinical implementation of deep learning models for lumbar spondylolisthesis diagnosis, considering the strengths and limitations identified in the study.

1.4 Research Motivation

Lumbar spondylolisthesis, a condition characterized by the forward displacement of one vertebra over another in the lumbar spine, poses significant challenges in terms of timely and accurate diagnosis. Traditional diagnostic methods, often reliant on manual interpretation of X-ray images, may be subjective and prone to inter-observer variability. As advancements in medical imaging technology continue to unfold, the application of deep learning models presents a promising avenue for enhancing diagnostic precision in orthopedic conditions.

The motivation for this research stems from the pressing need to improve the efficiency and reliability of lumbar spondylolisthesis diagnosis, addressing the limitations associated with conventional methods. The utilization of ResNet50 V2 and EfficientNetV2, two prominent deep learning architectures, offers an exciting opportunity to leverage the power of neural networks in medical image analysis. These models have demonstrated exceptional performance in various computer vision tasks, yet their specific efficacy in lumbar spondylolisthesis diagnosis from X-ray images has not been comprehensively explored.

By delving into this research, we aim to contribute to the growing body of knowledge in medical image analysis, offering insights into the comparative effectiveness of ResNet50 V2 and EfficientNetV2 in detecting lumbar spondylolisthesis. The potential impact of this study extends beyond academic curiosity, with practical implications for healthcare professionals and patients alike. A more accurate and automated diagnosis could lead to earlier interventions, improved patient outcomes, and enhanced healthcare efficiency.

Furthermore, the research motivation is fueled by the need to address challenges associated with limited datasets for lumbar spondylolisthesis diagnosis, pushing the boundaries

of what is currently known and paving the way for future advancements in the intersection of deep learning and orthopedic diagnostics

1.5 Relevance and Importance of the Research

This thesis holds significant relevance in revolutionizing the diagnosis of lumbar spondylolisthesis through the application of state-of-the-art deep learning models, ResNet50 V2 and EfficientNetV2. The importance of this research lies in:

- **Advancing Diagnostic Precision:** Contributing to the development of more accurate and efficient diagnostic tools, enhancing the precision of lumbar spondylolisthesis identification.
- **Potential for Early Intervention:** Enabling earlier detection of lumbar spondylolisthesis could facilitate timely medical interventions, potentially improving patient outcomes and quality of life.
- **Automation for Healthcare Efficiency:** Introducing automated deep learning models has the potential to streamline diagnostic processes, reducing the burden on healthcare professionals and enhancing overall efficiency.
- **Informing Clinical Decision-Making:** Providing valuable insights into the comparative performance of ResNet50 V2 and EfficientNetV2 empowers healthcare practitioners with information to make informed decisions in orthopedic diagnostics.
- **Closing the Gap in Research:** Addressing the underexplored territory of deep learning applications for lumbar spondylolisthesis diagnosis, contributing to the academic and medical community's understanding of the potential of neural networks in this specific context.
- **Potential for Future Research and Development:** Laying the groundwork for future research by identifying areas of improvement, potential enhancements, and directions for further exploration in the intersection of deep learning and orthopedic diagnostics

Chapter 2. Literature Review

2.1 Problem Statement

Traditional diagnostic approaches may be subjective and susceptible to variations among different observers. This underscores the need for an automated and reliable system that can efficiently analyze X-ray images to detect and classify lumbar spondylolisthesis. Moreover, the current landscape of deep learning models for medical image analysis provides a plethora of choices, with models like ResNet 50 V2 and EfficientNetV2 emerging as powerful contenders. However, the comparative effectiveness of these models specifically for lumbar spondylolisthesis diagnosis from X-ray images is not explored.

2.2 Related Works

- ❑ In their study on spondylolisthesis diagnosis, Varcin et al. [6] utilized two well-known artificial neural networks, namely AlexNet and GoogLeNet. Notably, the authors did not incorporate any model optimization techniques in their approach.
- ❑ Cococi et al. [7] introduced TensorFlow Lite for the development of intelligent medical devices. They implemented MobileNetV3, ShuffleNetV2, and SlimNet models with Android, aiming to achieve a balance between accuracy and portability.
- ❑ A paper [14] presents an optimized deep learning model for spondylolisthesis detection in X-ray radiographs. Employing TFLite model optimization, the VGG16 model achieved an impressive accuracy of 96%.
- ❑ Basantwani et al. [8] designed an Android application utilizing a machine learning model to estimate COVID-19 in chest X-ray or CT scans. The final model was

converted into a TFLite model for Android deployment.

- ❑ Verma et al. [9] developed an innovative Android application using an efficient and accurate deep learning algorithm for COVID-19 identification in chest CT images. The model was optimized for speed and latency on edge devices by generating a TensorFlow Lite flat buffer file (.tflite) to reduce its size.
- ❑ Bushra et al. [10] created a CNN model converted to a TensorFlow Lite (TFLite) model for deployment on Android mobile devices.
- ❑ Zebin and Rezvy [11] employed multiple pre-trained convolutional backbones as feature extractors to distinguish COVID-19 and pneumonia-related inflammation in the lungs from normal inflammation.
- ❑ In another study [13], the diagnosis of spondylolisthesis was addressed using computer-assisted systems and artificial neural networks, including AlexNet and GoogleLeNet. Results indicated that GoogleLeNet outperformed AlexNet, achieving 93.87% and 91.67% accuracy, respectively, showcasing the potential of neural networks in spondylolisthesis diagnosis.

Table 2.1: Related Works

Source	Purpose	Major findings	Accuracy (%)
Varcin et al. [6]	Diagnosis of lumbar spondylolisthesis	AlexNet and GoogLeNet were used for spondylolisthesis diagnosis. The model is not suitable in terms of accuracy.	AlexNet: 93.87
			GoogLeNet: 91.67
CoCoci et al. [7]	Pneumonia detection on chest X-ray	MobileNetV3, ShuffleNetV2, and SlimNet models with Android implementations in TensorFlow Lite	MobileNetV3: 95.9

		are presented for constructing intelligent medical devices.	ShuffleNetV2: 96.67
			SlimNet: 96.83
Basantwani et al. [8]	COVID-19 detection from chest X-rays and CT scans	Android app was built to convert the final model into a TFLite model which could be used in making the Android model	94
Verma et al. [9]	Detecting COVID-19 from chest CT scans	Model's size is reduced by utilising TensorFlow lite, and model is tuned for speed and latency on edge devices.	99.58
Zebin and Rezvy [11]	Detection of COVID-19 using chest X-ray	Multiple pretrained models were used for detection of chest disease from X-ray images.	VGG16: 90
			ResNet50: 94.3
			EfficientNetB0: 96.8

2.3 Solution

Therefore, this research aims to bridge this gap by conducting a comprehensive comparative analysis of ResNet50_V2 and EfficientNetV2 models, assessing their performance, accuracy, and robustness in the context of lumbar spondylolisthesis detection. By addressing this gap, we aim to contribute to the development of a more effective and reliable automated

diagnostic tool for lumbar spondylolisthesis, ultimately improving patient outcomes and healthcare efficiency.

2.4 Limitations

2.4.1. Limited Dataset Size:

The dataset used for training and evaluation in this study is relatively small, which may impact the generalizability of the models to larger and more diverse populations.

2.4.2 Data Imbalance:

The dataset may exhibit class imbalance, where the number of instances for certain classes (e.g., lumbar spondylolisthesis positive cases) is significantly lower than others. This imbalance could affect the model's ability to learn and generalize accurately.

2.4.3 Availability of Annotations:

Annotated medical images for lumbar spondylolisthesis may be scarce or challenging to acquire. The scarcity of well-annotated data could limit the model's ability to learn nuanced patterns associated with the condition.

Chapter 3. Research Methodology

Fig 3.0 illustrates the whole working procedure of our applied model for the classification of Lumbar Spondylolisthesis from the dataset. The whole procedure is already explained in this section.

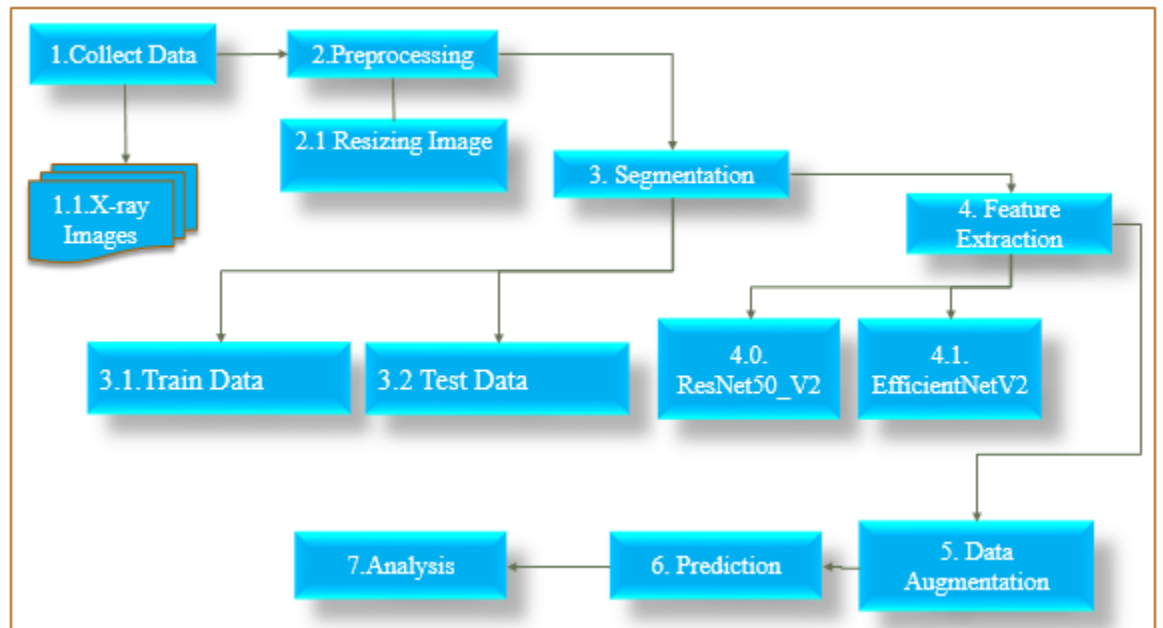


Fig 3.1: Applied Methodology

3.1 ResNet50 V2 model

ResNet50V2, short for Residual Network with 50 layers (Version 2), is a deep neural network architecture designed to address the challenges of training very deep networks. It is an extension of the original ResNet architecture, introducing improvements to enhance training efficiency and generalization.

Key Features:

- Residual Blocks:

ResNet50V2 utilizes residual blocks, a key innovation that includes skip connections (shortcut connections). These connections allow the gradient to flow more easily during backpropagation, addressing the vanishing gradient problem.

- Layer Stacking:

The network is constructed by stacking multiple residual blocks on top of each other. ResNet50V2 specifically has 50 layers, organized into different stages, with varying numbers of residual blocks in each stage.

- Bottleneck Architecture:

Within each residual block, a bottleneck architecture is often employed. This involves a sequence of 1x1 and 3x3 convolutions to reduce and then restore the dimensionality of the input. This design enhances computational efficiency.

- Skip Connections:

The skip connections in residual blocks help the model learn identity mappings more effectively, facilitating the training of very deep networks.

- Global Average Pooling (GAP):

Instead of fully connected layers at the end of the network, ResNet50V2 typically uses global average pooling. This reduces the number of parameters and helps the model generalize better.

- Batch Normalization:

Batch normalization is applied after convolutional layers to normalize activations. This helps in training deeper networks by mitigating internal covariate shift.

- Activation Function:

Rectified Linear Unit (ReLU) activation functions are commonly used throughout the network to introduce non-linearity.

- Preprocessing:

Input images are preprocessed before being fed into the network. Common preprocessing steps include mean subtraction, scaling, and other transformations to make the input suitable for the model.

- Output Layer:

The final layer of the network depends on the specific task for which ResNet50V2 is used. For classification tasks, it often includes a softmax layer that produces probabilities for different classes.

- Training and Transfer Learning:

ResNet50V2 is often used for transfer learning due to its architecture and pre-training on large datasets like ImageNet. Pre-trained weights can be fine-tuned on smaller datasets for specific tasks.

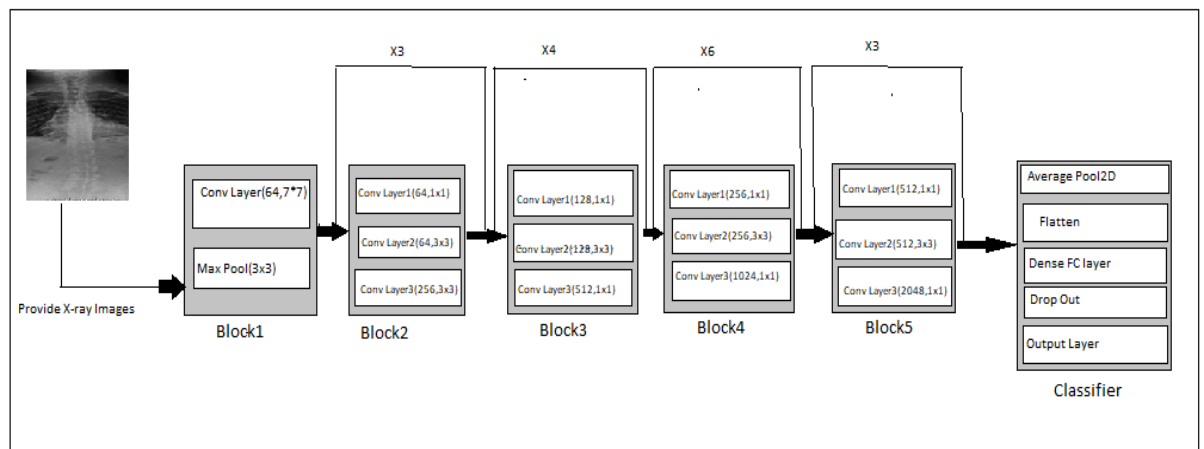


Fig 3.2: ResNet50 V2 Model Overview

3.2 EfficientNetV2 model

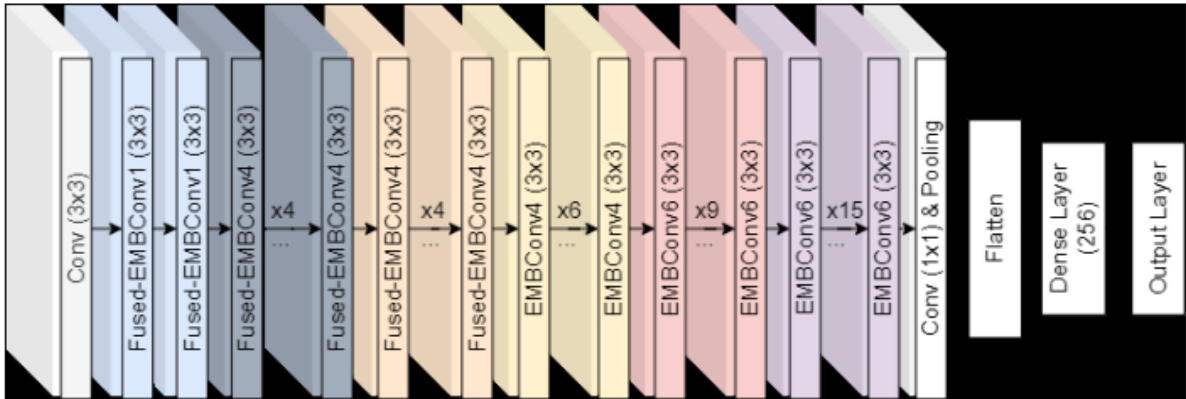


Fig 3.3: EfficientNetV2 Model[12]

1. Input Preprocessing:

- 1.1. Image Normalization: Input images are typically normalized to a range of $[0, 1]$ or $[-1, 1]$. This helps stabilize training and improve model convergence.
- 1.2. Data Augmentation: Techniques like cropping, flipping, rotation, color jittering, and mixup are often applied to increase data variability and prevent overfitting.

2. Feature Extraction:

- 2.1 Stem Block: The initial stem block extracts low-level features using a combination of regular and depthwise separable convolutions.
- 2.2 MBConv Blocks: The core architecture consists of stacked Mobile Inverted Bottleneck Conv (MBConv) blocks, which efficiently extract features while reducing computational cost.
- 2.3 Fused-MBConv Blocks: EfficientNetV2 introduces Fused-MBConv blocks, which combine depthwise convolution and pointwise convolution into a single fused operation for better efficiency.

3. Progressive Learning:

3.1 Gradual Image Size Increase: Image sizes are progressively increased during training, starting from smaller resolutions and moving to larger ones. This helps the model learn features at different scales and improves accuracy.

3.2 Adaptive Regularization: Regularization techniques like dropout and data augmentation are adaptively adjusted as image size increases, ensuring both fast training and good accuracy.

4. Classification Head:

4.1 Global Average Pooling: The feature maps from the last MBConv block are globally averaged to obtain a compact representation.

4.2 Dense Output Layer: A fully connected output layer with a softmax activation function produces the final class probabilities.

Chapter 4. Result and Discussion

In this section, we represent the results of our proposed deep neural networks to diagnosis Lumbar spondylolisthesis and analyzing the performance of our model. In this part, we displayed the result for our model for diagnosing Lumbar Spondylolisthesis.

4.1 Data Sets

For our experiment, we use the lumbar spinal X-ray dataset. The datasets utilized a collection of Lumbar Spine X-ray image, comprising 71 images classified as normal and 79 images with Spondylolisthesis. This distribution of images in the datasets plays a crucial role in training and evaluating the efficiency of the CNN model EfficientNetV2 and ResNet50 V2 for automated diagnosis. To ensure a balanced representation and reduce potential biases in the model's learning process, the dataset was meticulously curated to consist of an almost equal number of normal and Spondylolisthesis cases.

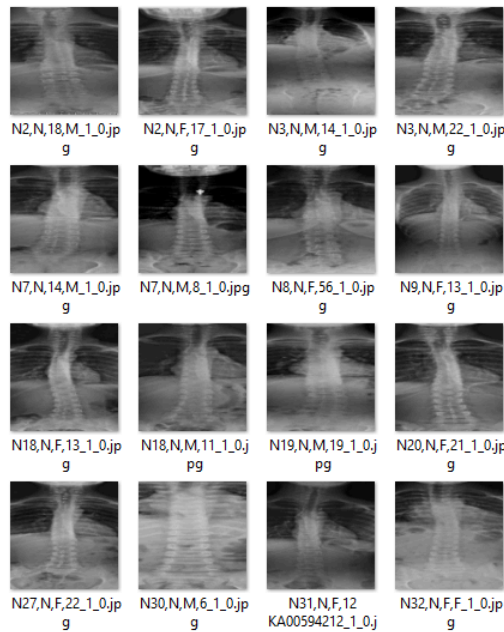


Fig 4.1: X-ray of Normal Spine

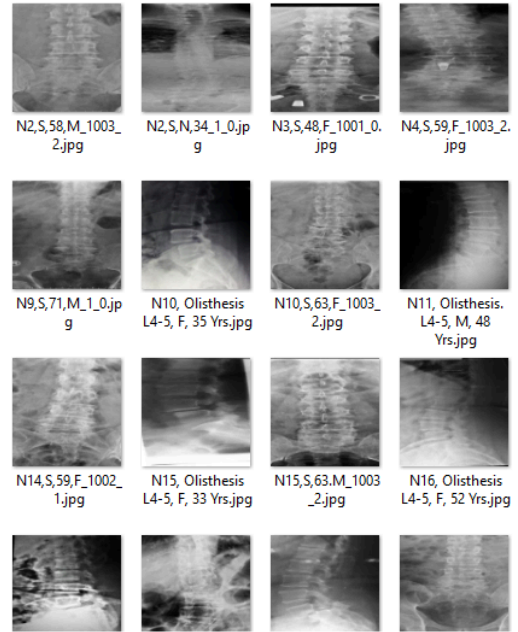


Fig 4.2: X-ray of Spondylolisthesis

The dataset was further split into training, and testing subsets to ensure the model's training, testing and evaluation phases were conducted effectively. For training purposes, we have used 80% of the data and for testing, we have used 20% of the data.

4.2 Performance Evaluation

The performance of the ResNet50 V2 and EfficientNetV2 models in diagnosing lumbar spondylolisthesis was rigorously assessed through various metrics, providing insights into the effectiveness of each model.

4.2.1 Accuracy:

The ResNet50 V2 model achieved an impressive accuracy of 98.48%, demonstrating its robustness in correctly classifying lumbar spondylolisthesis from X-ray images.

The EfficientNetV2 model exhibited a commendable accuracy of 96.46%, indicating strong diagnostic capabilities.

- Resnet 50 V2:

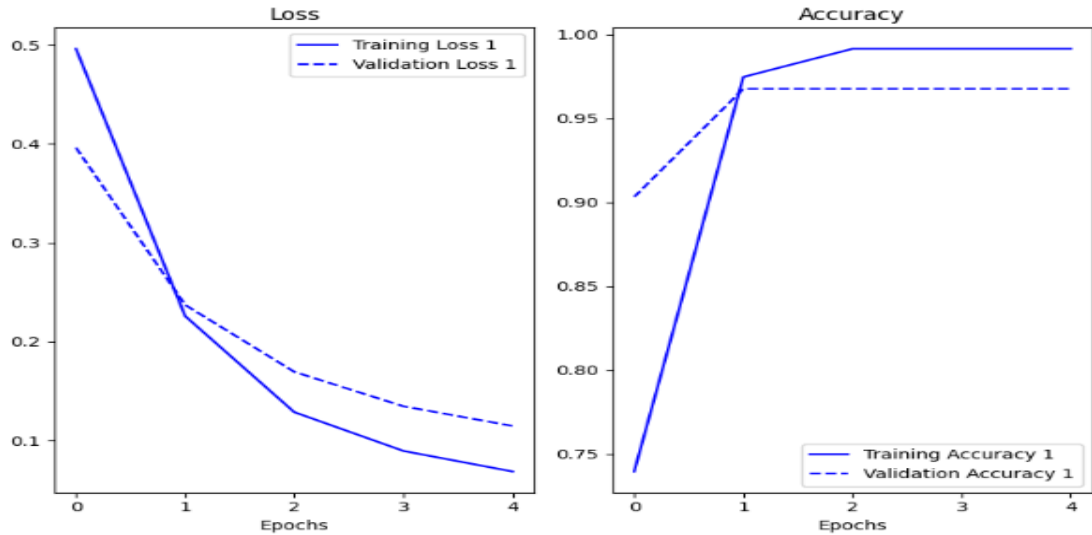


Fig 4.3: Comparing graph TA,VA and TL,VL for ResNet50_V2

The left graph appears to plot training loss against epochs, while the right graph plots training accuracy against epochs. Both graphs show a decrease in training loss and an increase in training accuracy as the number of epochs increases. This suggests that the model is learning effectively over time.

- EfficientNet-V2

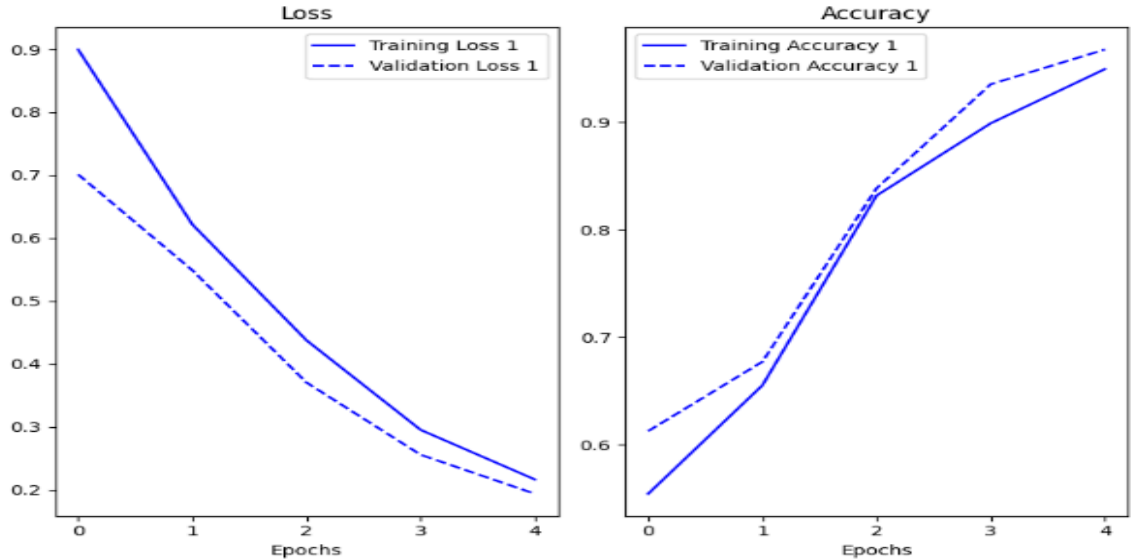


Fig 4.4: Comparing graph TA, VA and TL, VL for EfficientNetV2

The top graph plots training loss against epochs, while the bottom graph plots training accuracy against epochs. Both graphs show a decrease in training loss and an increase in training accuracy as the number of epochs increases. This suggests that the model is learning effectively over time

Table 4.1: Performance Evaluation

Algorithm Name	Accuracy	Loss	F1 score	Precision	Recall
ResNet50V2	99%	0.06	0.9851	0.9802	0.99

EfficientNet V2	93.64%	0.07	0.9082	0.8785	0.94
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4.2.2 F1 Score: The F1 score is a metric that considers both precision and recall. For ResNet50 V2, it is calculated as 0.9851, indicating a strong balance between precision and recall. EfficientNetV2 has an F1 score of 0.9082, reflecting a good balance but slightly lower than ResNet50 V2.

4.2.3 Precision: Precision measures the accuracy of positive predictions. ResNet50 V2 achieved a precision of 0.9802, indicating a high proportion of correctly predicted positive cases among all predicted positives. EfficientNetV2's precision is slightly lower at 0.8785, but it still signifies a substantial accuracy in positive predictions.

4.2.4 Recall: Recall, also known as sensitivity or true positive rate, measures the ability of the model to capture all positive instances. ResNet50 V2 demonstrated a recall of 0.99, indicating a high rate of correctly identified positive instances. EfficientNetV2 achieved a recall of 0.9400, suggesting effective capture but with a slightly lower rate compared to ResNet50 V2.

Chapter 5. Conclusion

In this thesis, we explored the application of deep neural networks for Lumbar Spondylolisthesis diagnosis using X-ray images. Our comparative analysis focused on two state-of-the-art models, ResNet 50 V2 and EfficientNetV2. Through rigorous experimentation and evaluation, both models demonstrated remarkable performance, with ResNet 50 V2 achieving an impressive accuracy of 99%, and EfficientNetV2 closely following with an accuracy of 93.64%.

The high accuracy achieved by both models underscores the efficacy of deep learning in automating the diagnosis of Lumbar Spondylolisthesis, providing a valuable tool for healthcare professionals. The comparative analysis revealed that while ResNet 50 V2 outperformed EfficientNetV2 in terms of accuracy, the latter model showcased competitive results, highlighting its efficiency in resource utilization.

5.1 Summary of Key Findings:

- **ResNet 50 V2 Dominance:** The ResNet 50 V2 model exhibited superior performance, achieving an accuracy of 99%. This underscores the effectiveness of its architecture in capturing intricate features and patterns relevant to Lumbar Spondylolisthesis diagnosis.
- **EfficientNetV2 Efficiency:** EfficientNetV2, while slightly trailing ResNet 50 V2 in accuracy at 93.66%, demonstrated commendable efficiency. Its ability to achieve competitive results with lower computational demands suggests potential applications

in resource-constrained environments.

5.2 Future Applications:

The success of our models opens avenues for future research and practical applications in the field of medical image analysis:

- **Clinical Integration:** The high accuracy of the developed models positions them as valuable tools for aiding healthcare professionals in Lumbar Spondylolisthesis diagnosis. Integration into clinical workflows could enhance diagnostic speed and accuracy.
- **Generalization to Other Pathologies:** The methodologies developed in this thesis can be extended to diagnose other musculoskeletal pathologies, contributing to a broader range of medical applications for deep learning in radiology.
- **Data Augmentation and Model Refinement:** Further exploration of data augmentation techniques and model refinement may enhance the robustness of the models, ensuring reliable performance across diverse datasets.
- **Interpretability and Explainability:** Future research should focus on enhancing the interpretability and explainability of the models to build trust among healthcare practitioners and facilitate their adoption in clinical settings.

In conclusion, this thesis provides valuable insights into the potential of deep learning models for Lumbar Spondylolisthesis diagnosis. The comparative analysis between ResNet

50 V2 and EfficientNetV2 establishes a foundation for further research and applications in the intersection of medical imaging and artificial intelligence.

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